

```
from sympy import *
```

## Ring of integers of a cubic field

We can use `sympy` to work in the ring of integers of the field  $\mathbb{Q}[\alpha]$  where  $\alpha$  is a root of  $P(X) = X^3 + 3X + 1$ .

```
x = Symbol('x')
pol = x**3 + 3*x + 1
```

We can define this as a ring in `sympy`.

```
from sympy.polys.agca.extensions import FiniteExtension
R = FiniteExtension(Poly(x**3 + 3*x + 1)); R
ZZ[x]/(x**3 + 3*x + 1)
```

Alternatively, we can work with matrices.

The following is the companion matrix of  $P(X) = X^3 + 3X + 1$ .

```
alpha = ImmutableMatrix([[0,1,0],[0,0,1],[-1,-3,0]]);alpha
```

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -3 & 0 \end{bmatrix}$$

## Known Details

The following results have been proved in a different worksheet.

- The discriminant of  $X^3 + X + 1$  is  $-135$ .
- $\mathbb{R} \otimes_{\mathbb{Z}} R$  is isomorphic to  $\mathbb{R} \times \mathbb{C}$  since the discriminant is negative so two roots are complex.
- $R_3 = \mathbb{Z}_3[\alpha] = \mathbb{Z}_3 \otimes_{\mathbb{Z}} R$  is a ramified extension of  $\mathbb{Z}_3$  of degree 3.
- $\mathbb{Z}_5[\alpha] = \mathbb{Z}_5 \otimes_{\mathbb{Z}} R$  is a product of  $\mathbb{Z}_5$  and  $S_5$  where  $S_5$  is a ramified extension of  $\mathbb{Z}_5$  of degree 2.
- For all primes  $p \neq 3, 5$ , the base change  $\mathbb{Z}_p \otimes_{\mathbb{Z}} R$  is a product of unramified extensions of  $\mathbb{Z}_p$ .
- $R$  is the ring of integers in the field  $K[\alpha] = K[X]/\langle X^3 + 3X + 1 \rangle$ .

It follows that to determine the prime ideals of  $R$ , we need to:

- Determine the ideals lying over 3 and 5.
- Determine the ideals lying over unramified primes.

## Basic structure of $R$

This matrix  $\alpha$  is invertible since it is a unit in the ring of algebraic integers.

```
alphai = alpha.inverse(); alphai
```

$$\begin{bmatrix} -3 & 0 & -1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

We can directly check this in the ring  $R$ .

```
R(x)*(-3-x**2)
```

```
1
```

In order to work with matrices we need the basis  $\{1, \alpha, \alpha^2\}$ .

```
alpha2 = alpha**2; alpha2
```

$$\begin{bmatrix} 0 & 0 & 1 \\ -1 & -3 & 0 \\ 0 & -1 & -3 \end{bmatrix}$$

A funny way to get to the Identity matrix which is the other basis element.

```
id = -(alpha**3+3*alpha); id
```

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

We write a general element of  $R$ .

```
var('p,q,r')
```

```
gen = p*alpha2 + q*alpha+r*id; gen
```

$$\begin{bmatrix} r & q & p \\ -p & -3p+r & q \\ -q & -p-3q & -3p+r \end{bmatrix}$$

This is the norm form for  $R$ .

```
norm=gen.det(); norm
```

$$p^3 - 3p^2q + 9p^2r + 3pqr - 6pr^2 - q^3 + 3q^2r + r^3$$

We note that the norm of  $\alpha$  is  $\pm 1$  which confirms it is a unit!

```
norm.subs({p:0,q:1,r:0})
```

```
-1
```

### The prime 3

We have seen that  $P(X)$  becomes  $(X+1)^3$  modulo 3 and in terms of  $Y = X+1$  this is an Eisenstein polynomial.

```
y = Symbol('y')
```

```
epol = expand(pol.subs(x,y-1)); epol
```

$$y^3 - 3y^2 + 6y - 3$$

Thus, the image  $\alpha + 1$  of  $y$  in  $R$  generates the prime ideal in  $R_3$ .

In fact, we see that the norm of  $\alpha + 1$  is 3. This means that  $R/\langle \alpha + 1 \rangle$  is  $\mathbb{F}_3$ . So  $\alpha + 1$  is a prime in  $R$ .

```
norm.subs({p:0,q:1,r:1})
```

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This means the 3 is a unit multiple of  $(\alpha + 1)^3$ .

We calculate the unit as  $\alpha^{-2}$ .

```
(R(x)+1)**3
```

```
3*x**2
```

```
pi3 = alpha+id
alphai**2*pi3**3
```

$$\begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

## The prime 5

We had seen that 1 is a simple root of  $P(X)$  in  $\mathbb{F}_5$  and  $-2$  is a double root.

So we note that  $\alpha - 1$  will become prime in the  $\mathbb{Z}_5$  factor of  $\mathbb{Z}_5[\alpha]$ .

Once again, when we calculate its norm, we see that  $\alpha - 1$  is a prime in  $R$ .

```
norm.subs({p:0,q:1,r:-1})
```

-5

```
pi5_1 = alpha-id
```

Next, we calculate the norm of  $\alpha - 2$ .

```
norm.subs({p:0,q:1,r:-2})
```

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We see that this is not a prime.

However, we see that it *must* be divisible by  $\alpha + 1$ . So we calculate  $(\alpha - 2)(\alpha + 1)^{-1}$ .

```
(alpha-2*id)*((alpha+id).inverse())
```

$$\begin{bmatrix} -3 & 1 & -1 \\ 1 & 0 & 1 \\ -1 & -2 & 0 \end{bmatrix}$$

This shows that  $-\alpha^2 + \alpha - 3$  is a prime

```
pi5_2 = -alpha2 + alpha - 3*id
pi5_2
```

$$\begin{bmatrix} -3 & 1 & -1 \\ 1 & 0 & 1 \\ -1 & -2 & 0 \end{bmatrix}$$

```
pi5_2.det()
```

```
-5
```

We note that 5 must be a unit multiple of the  $\pi_{5,1}$  and  $\pi_{5,2}^2$ .

```
(pi5_1*pi5_2**2)/5
```

$$\begin{bmatrix} -3 & 0 & -1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

Thus, the unit is  $-(\alpha^2 + 3) = \alpha^{-1}$ .

In other words,  $5 = \alpha \cdot \pi_{5,1} \cdot \pi_{5,2}^2$ .

## The prime 2

We say that  $P(X)$  is irreducible in  $\mathbb{F}_2[X]$ .

Thus, 2 is a prime in  $R$ .

This means that the only powers of 2 that are values of the norm form are powers of 8!

## The prime 7

We saw that  $-3$  is a root of  $P(X)$  in  $\mathbb{F}_7$ .

So we calculate the norm of  $\alpha + 3$ .

```
norm.subs({p:0,q:1,r:3})
```

```
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```

Thus, we see that  $\alpha + 3$  is *not* a prime.

However, it is either divisible by  $\pi_{5,1}$  or  $\pi_{5,2}$ .

```
(alpha+3*id)*pi5_1.inverse(), (alpha+3*id)*pi5_2.inverse()
```

```
(Matrix([
[-11/5, -4/5, -4/5],
[ 4/5,  1/5, -4/5],
[ 4/5, 16/5,  1/5]]),
Matrix([
[-1, -1, -1],
```

```
[ 1,  2, -1],
[ 1,  4,  2]])
```

Thus, we see that  $\alpha + 3$  is  $\pi_{5,2}(-\alpha^2 - \alpha - 1)$ .

We put  $\pi_{7,1} = \alpha^2 + \alpha + 1$  and check that it is a prime lying over 7.

```
pi7_1 = alpha2 + alpha + id
pi7_1.det()
```

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Since the other factor of  $\mathbb{Z}_7[\alpha]$  is unramified it is associated with a prime  $\pi_{7,2}$  such that 7 is a unit multiple of  $\pi_{7,1}$  and  $\pi_{7,2}$ .

We calculate  $7/\pi_{7,1}$ .

```
7*(pi7_1.inverse())
```

```
[ 8  -2   3 ]
[-3  -1  -2 ]
[ 2   3  -1 ]
```

We see that  $\pi_{7,2} = 3\alpha^2 - 2\alpha + 8$ .

```
pi7_2 = 3*alpha2 - 2*alpha + 8*id
pi7_2.det()
```

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Note that  $R/\langle\pi_{7,2}\rangle$  is  $\mathbb{F}_{7^2}$ , the quadratic extension of  $\mathbb{F}_7$ .

By construction  $7 = \pi_{7,1} \cdot \pi_{7,2}$ .

## Other primes

Now suppose that  $p$  is a prime different from 3 and 5.

Recall that to obtain the splitting of  $\mathbb{Z}_p[\alpha]$  it is enough to obtain the splitting of  $\mathbb{F}_p[X]/\langle P(X) \rangle$ . For this it is enough to check:

- Whether the discriminant  $(-135)$  of  $P(X)$  is a square modulo  $p$ .
- Whether  $P(X)$  has a root modulo  $p$ .

To check that former we need only calculate  $(-15)^{(p-1)/2}$  in  $\mathbb{F}_p$  (since  $-135 = 9 \cdot (-15)$ ).

To check the latter (except for  $p = 2$ !) we calculate the Euclidean division of  $X^p - X$  by  $P(X)$ .

- If  $X^p - X$  is divisible by  $P(X)$  in  $\mathbb{F}_p$ , then all three roots are in  $\mathbb{F}_p$ .
- Otherwise, we will get a quadratic remainder term. If  $P(X)$  has a root in  $\mathbb{F}_p$ , this quadratic remainder term will have a common linear factor with  $P(X)$  in  $\mathbb{F}_p$ .

For example, we can check this in the case  $p = 7$ .

```
gcd(Poly(x**7-x,domain=GF(7)),Poly(pol,domain=GF(7)))
```

```
Poly(x + 3, x, domain = F7)
```

This confirms (as we have seen above) that  $P(X)$  has exactly one root in  $\mathbb{F}_7$ .

```
for p in [11,13,17,19,23,29,31,37,41,47]:
    print(gcd(Poly(x**p-x,domain=GF(p)),Poly(pol,domain=GF(p))))

Poly(x - 4, x, modulus=11)
Poly(x + 2, x, modulus=13)
Poly(1, x, modulus=17)
Poly(1, x, modulus=19)
Poly(1, x, modulus=23)
Poly(x + 11, x, modulus=29)
Poly(1, x, modulus=31)
Poly(x - 3, x, modulus=37)
Poly(x + 12, x, modulus=41)
Poly(x**3 + 3*x + 1, x, modulus=47)
```

Thus, we see that  $P(X)$  has the following behaviour in  $\mathbb{F}_p$ .

- It has one root for  $p = 11, 13, 29, 37, 41$
- It has no root for  $p = 17, 19, 23, 31$
- It splits completely for  $p = 47$

A different approach to finding primes that split completely is as follows.

1. Look at primes  $p$  that divide the values of  $P(n)$  for various integers  $n$ .
2. For such a prime check that the discriminant  $(-135)$  is a square modulo  $p$ .